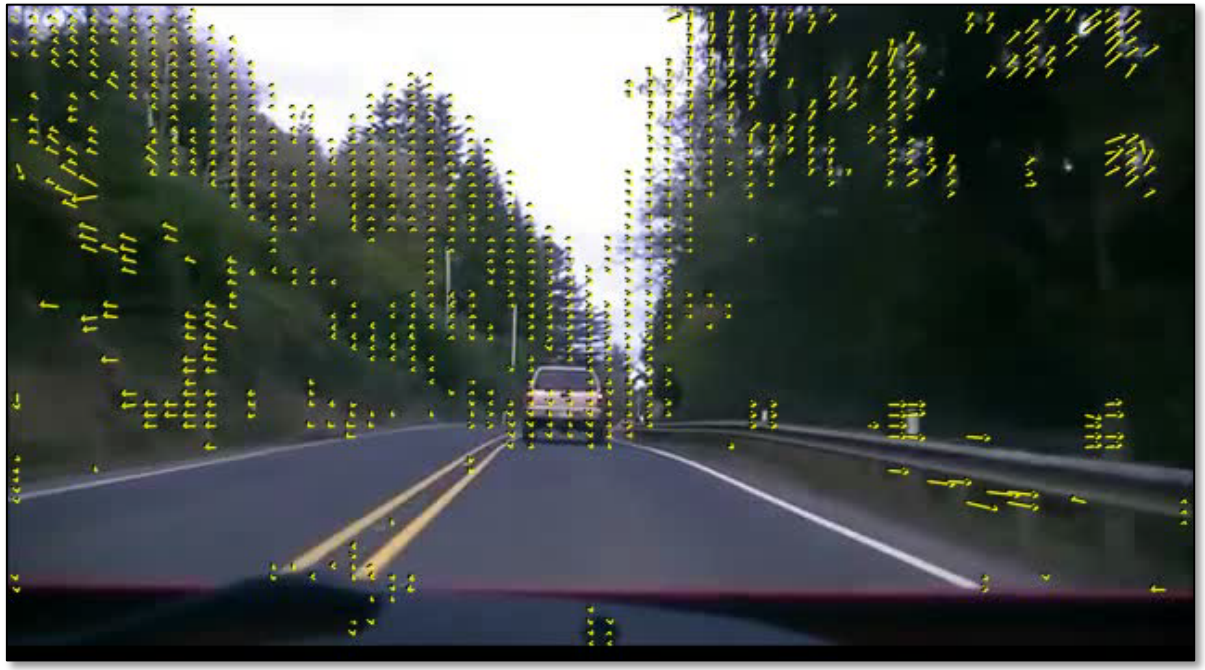
Displacement estimation

- Displacement estimation by block matching
  - Search strategies
  - Subpixel estimation



Where is the defect?

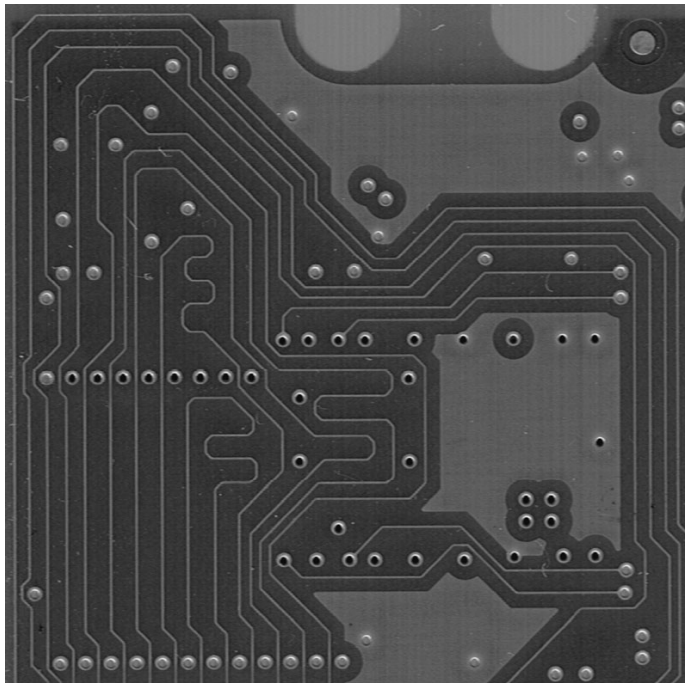


Image $g[x,y]$ (no defect)

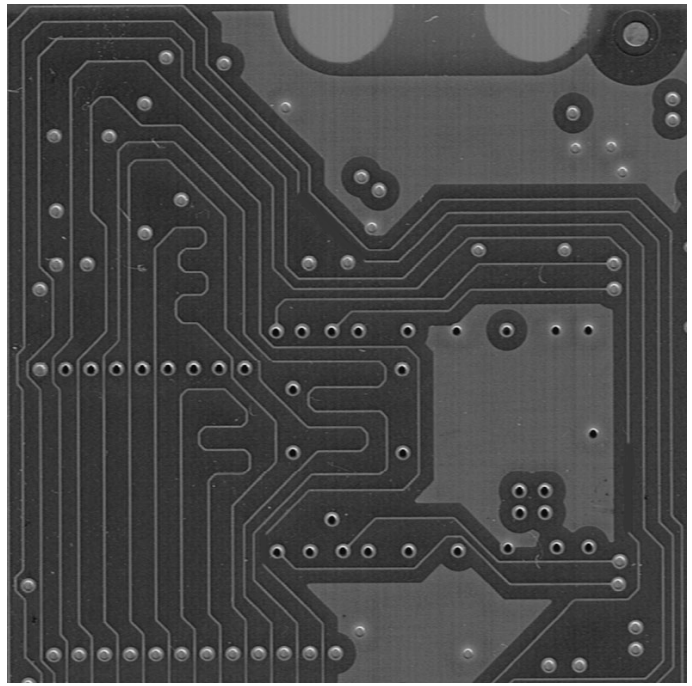
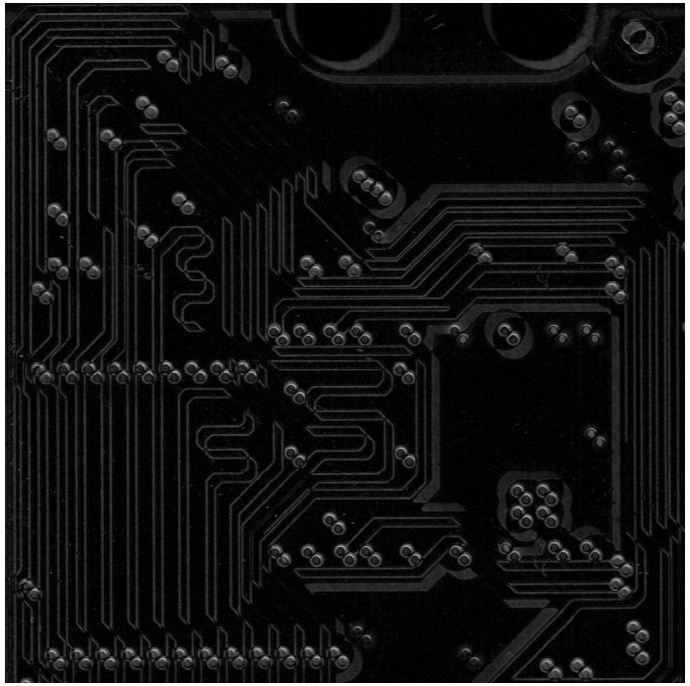
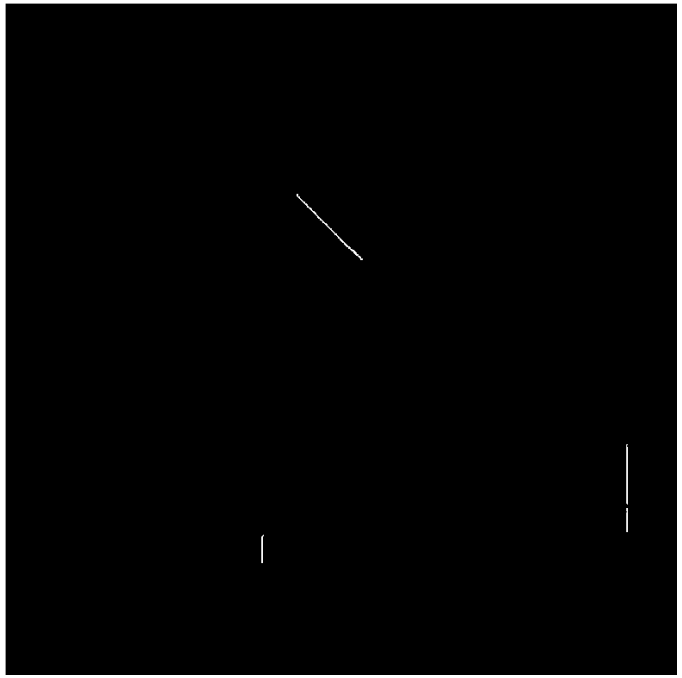


Image $f[x,y]$ (w/ defect)

Absolute difference between two images



$|f\text{-}g|$ w/o alignment



$|f\text{-}g|$ w/ alignment

Displacement estimation by block matching

Measurement window is compared with a shifted array of pixels in the other image, to determine the best match

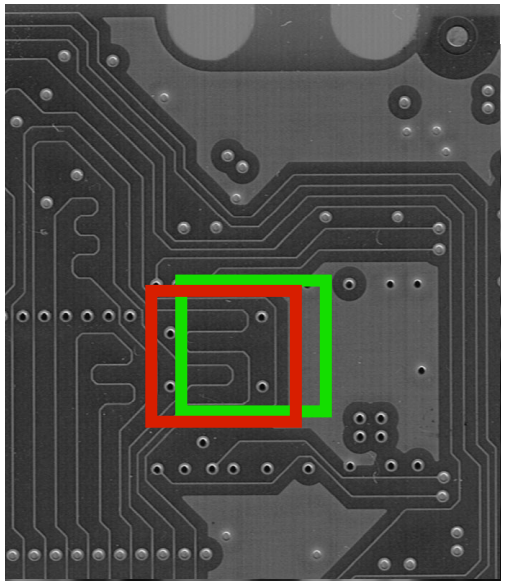


Image $g[x,y]$

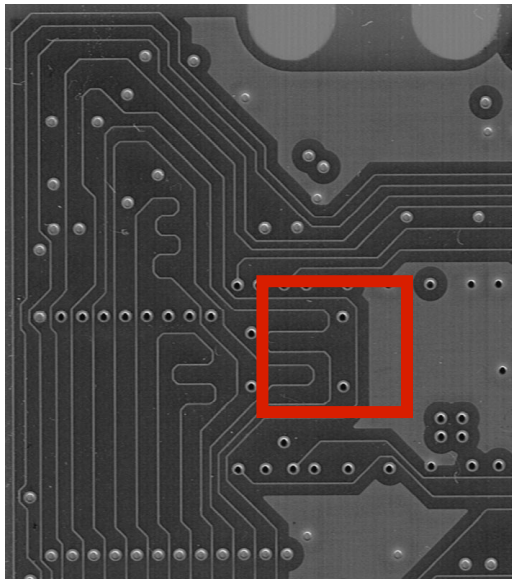


Image $f[x,y]$

Rectangular array of pixels is selected as a measurement window

Displacement estimation by block matching

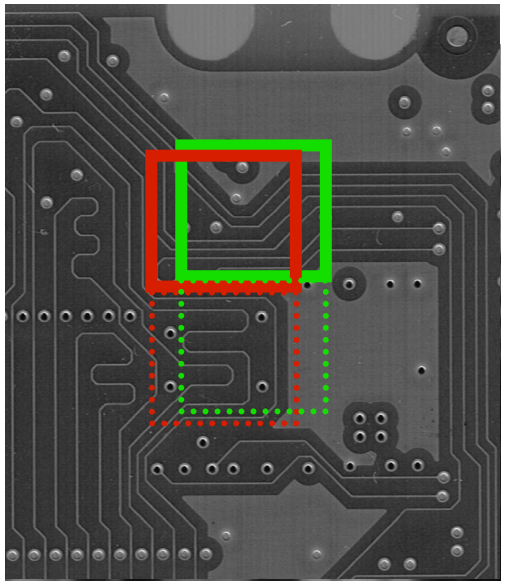


Image $g[x,y]$

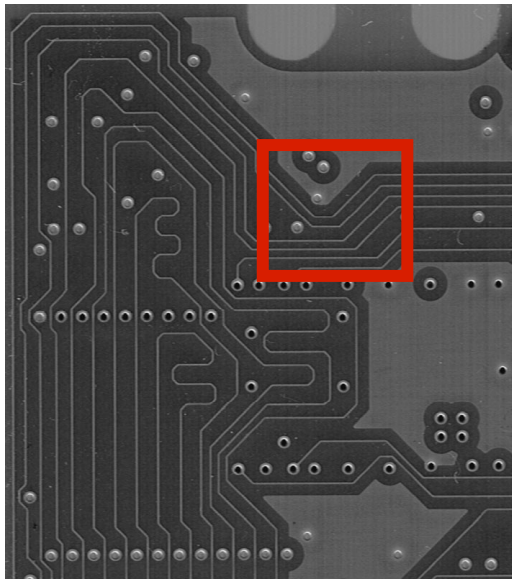


Image $f[x,y]$

. . . process repeated for another measurement window position.

Integer pixel shifts

Measurement window is compared with a shifted array of pixels in the other image, to determine the best match

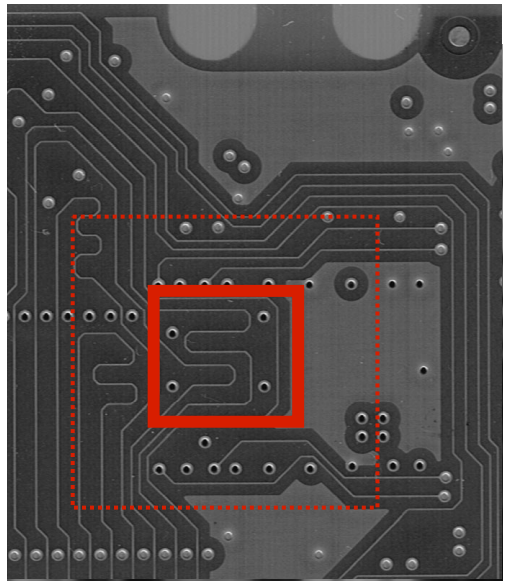


Image $g[x,y]$

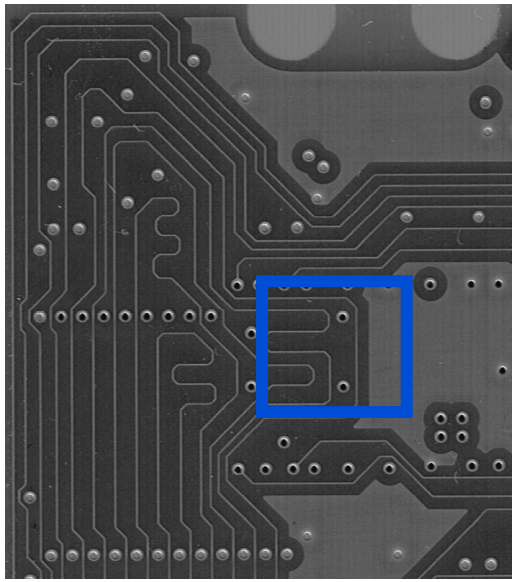
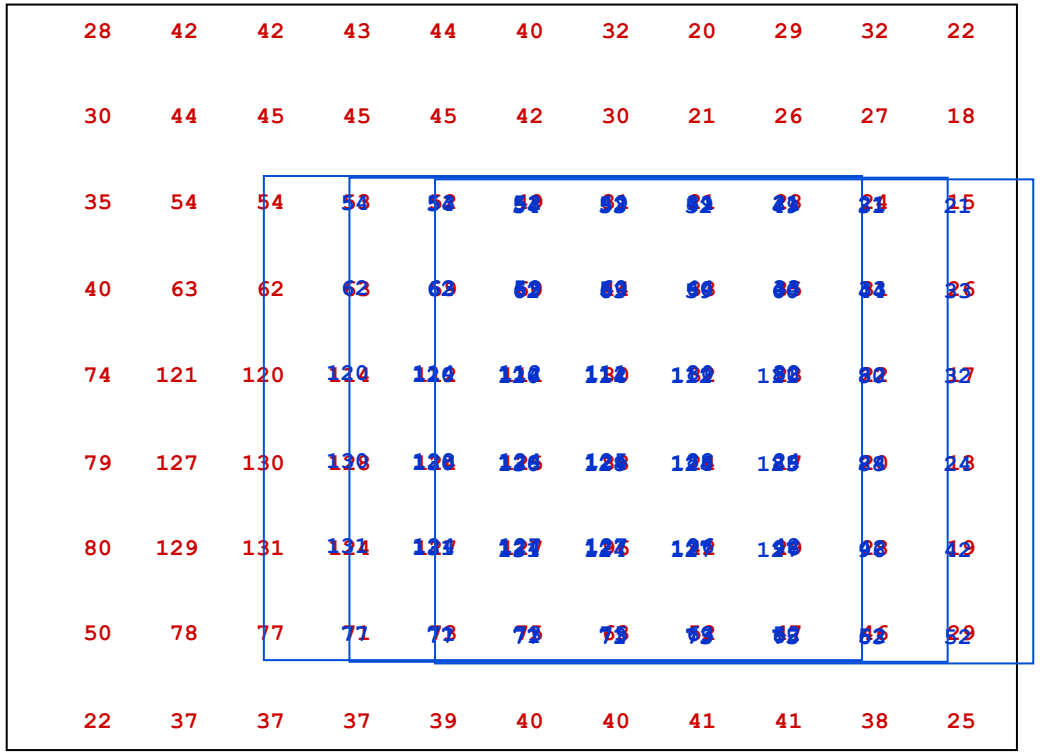


Image $f[x,y]$

Rectangular array of pixels is selected as a measurement window

Integer pixel shifts



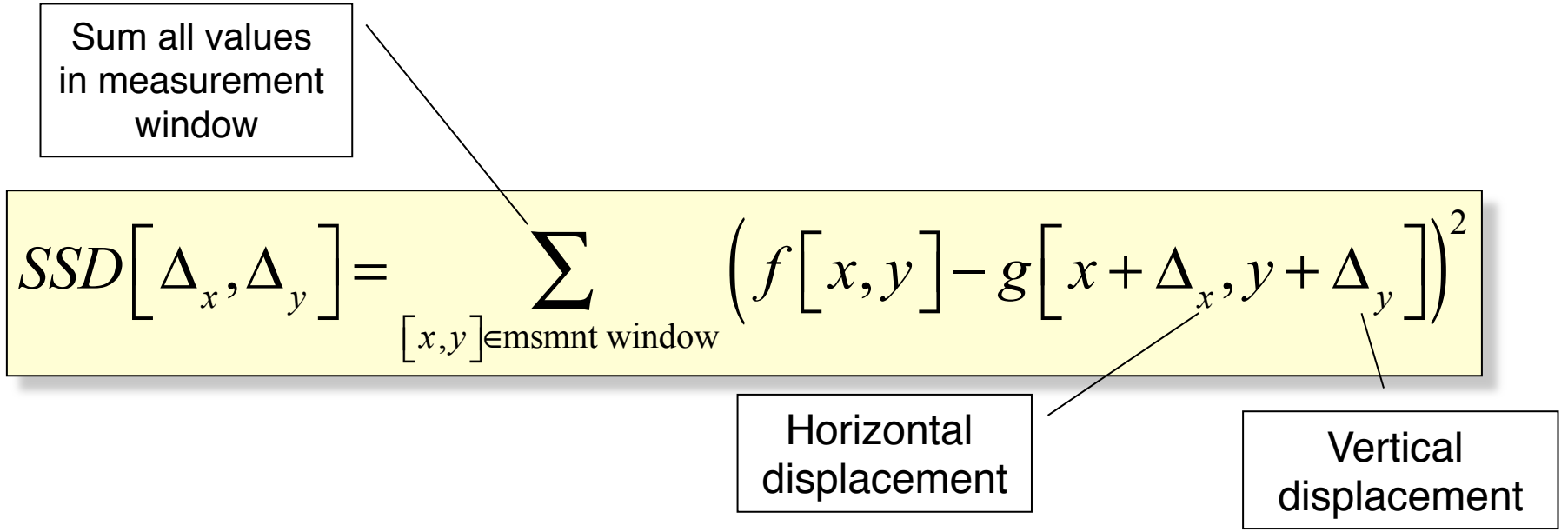
Measurement window is compared with a shifted array of pixels in the other image, to determine the best match

54	53	52	49	31	21
62	63	59	60	44	33
120	114	112	111	80	32
130	128	124	125	88	24
131	124	127	127	96	42
77	71	73	75	63	52

Rectangular array of pixels is selected as a measurement window

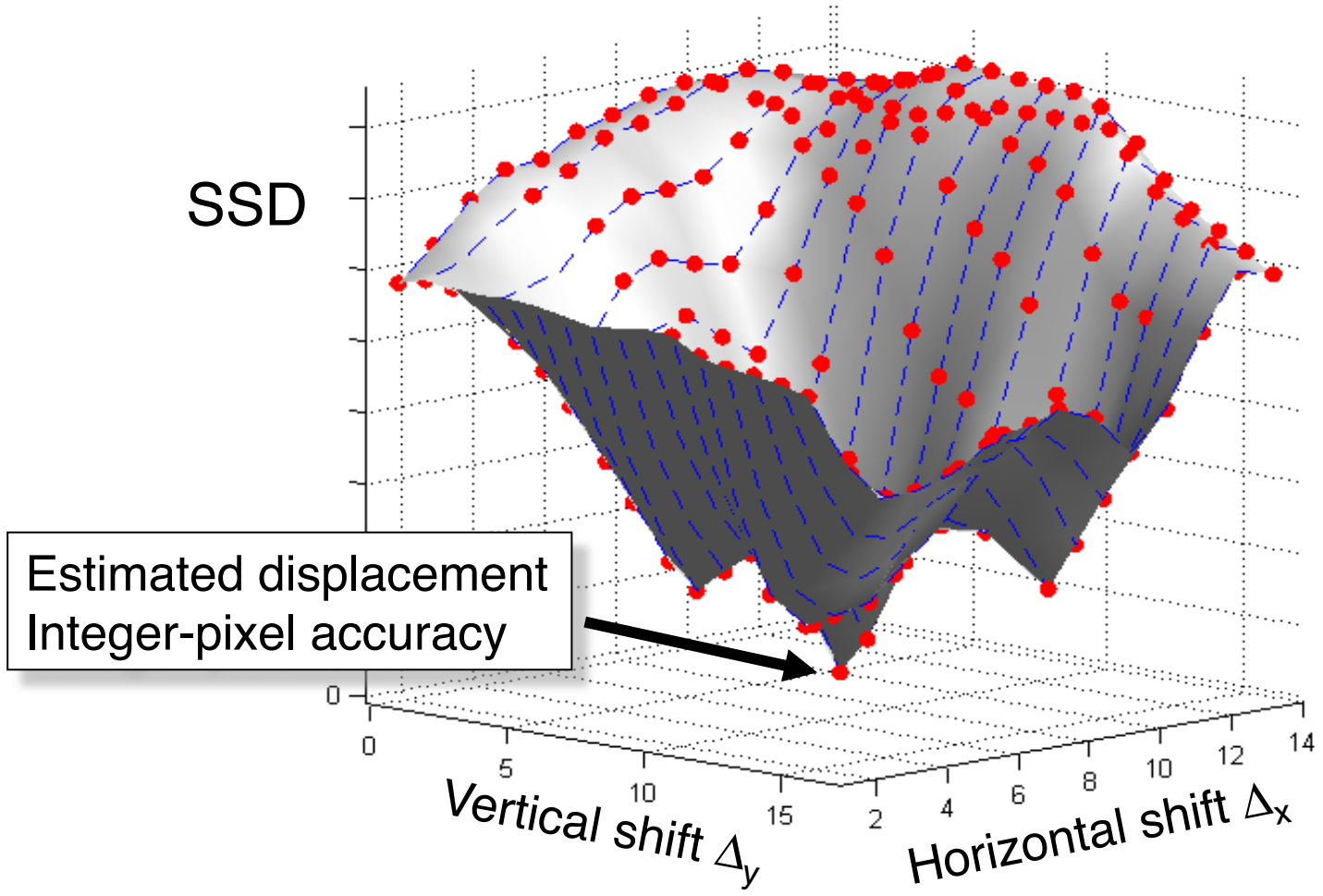
Error metric

- *Sum of Squared Differences*



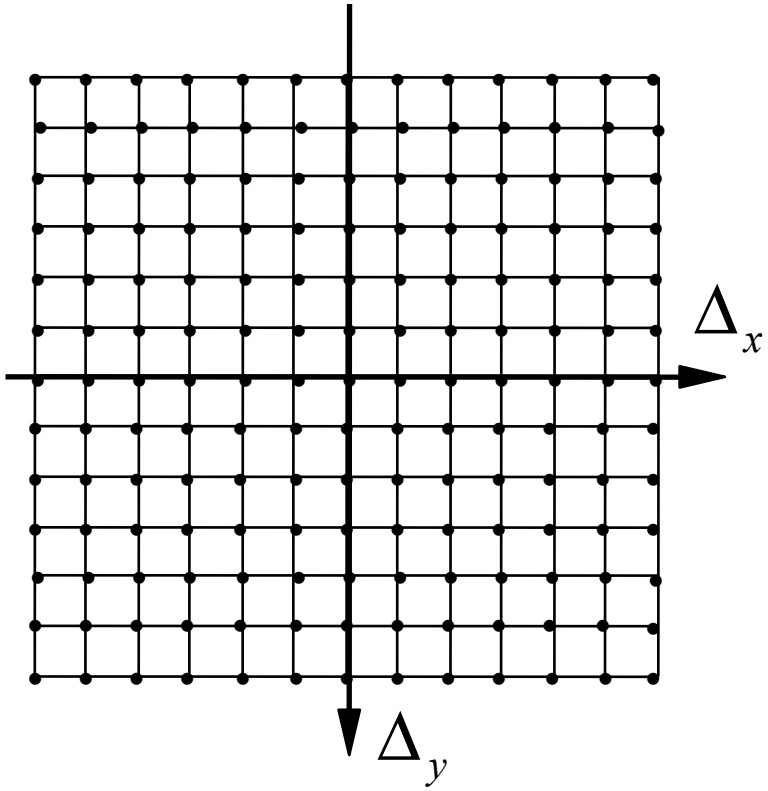
- Alternatives: SAD (*Sum of Absolute Differences*), cross correlation, mutual information . . .
- Robustness against outliers: sum of saturated squared differences, median of squared differences . . .

SSD values resulting from block matching



Block matching: search strategies

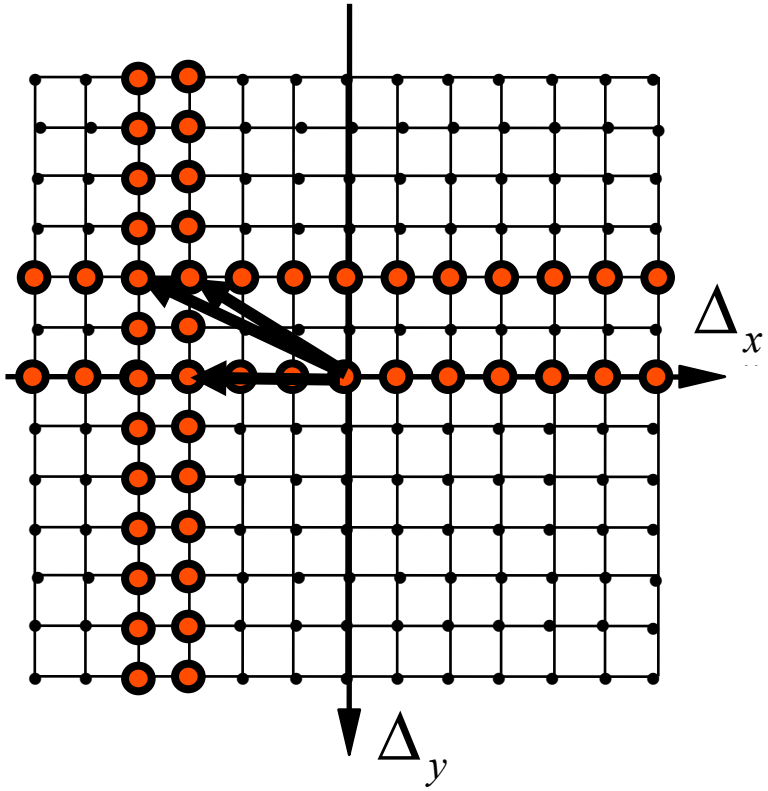
Full search



- All possible displacements within the search range are compared.
- Computationally expensive
- Highly regular, parallelizable

Block matching: search strategies

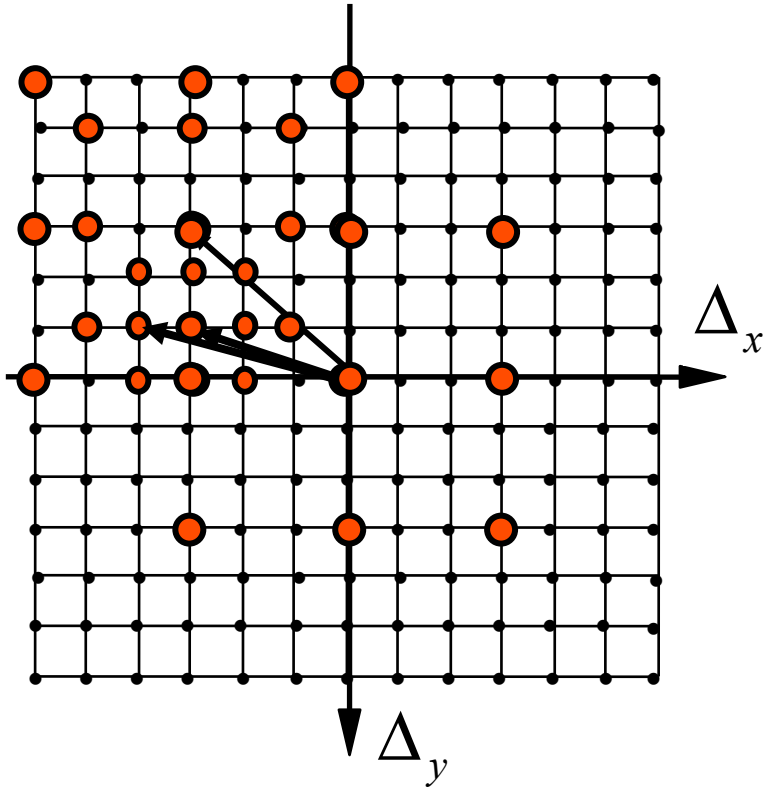
Conjugate direction search



- Alternate search in x and y directions
- Stop when there is no further improvement

Block matching: search strategies

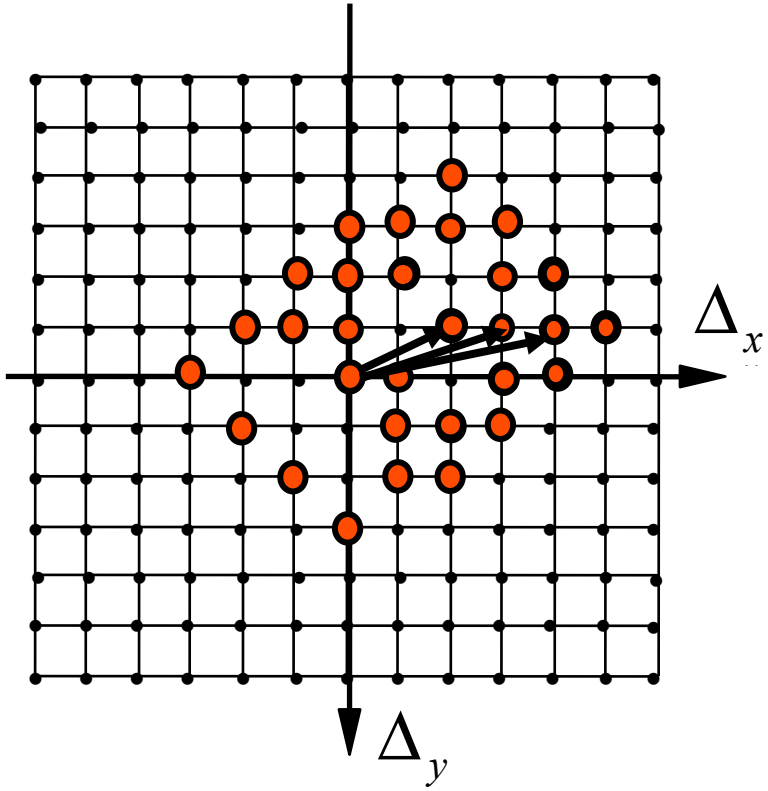
Coarse-to-fine



- Start with coarsely spaced candidate displacements
- Smaller pattern when best match is in the middle
- Stop when desired displacement accuracy is reached

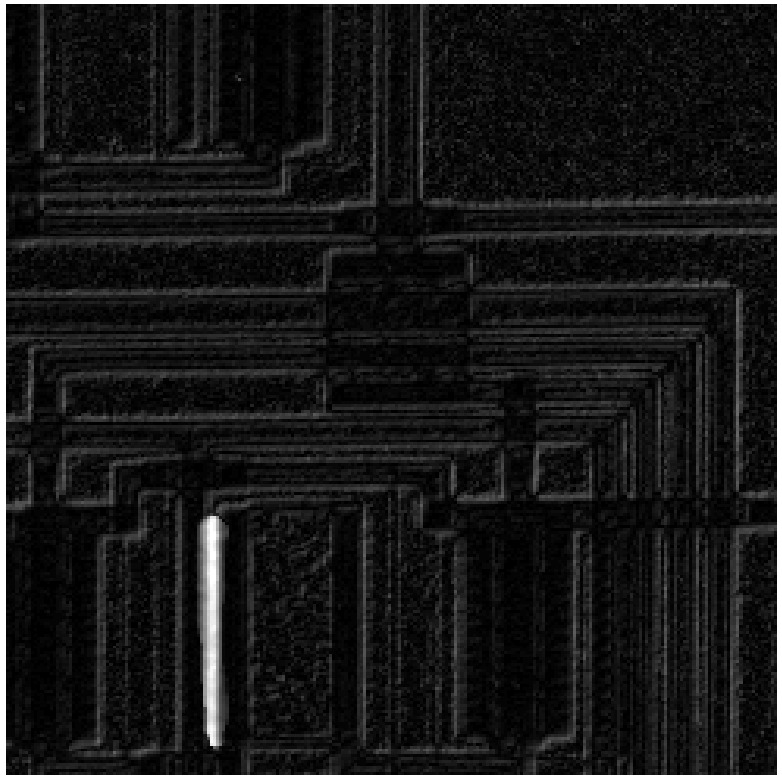
Block matching: search strategies

Diamond search [Li, Zeng, Liou, 1994] [Zhu, Ma, 1997]

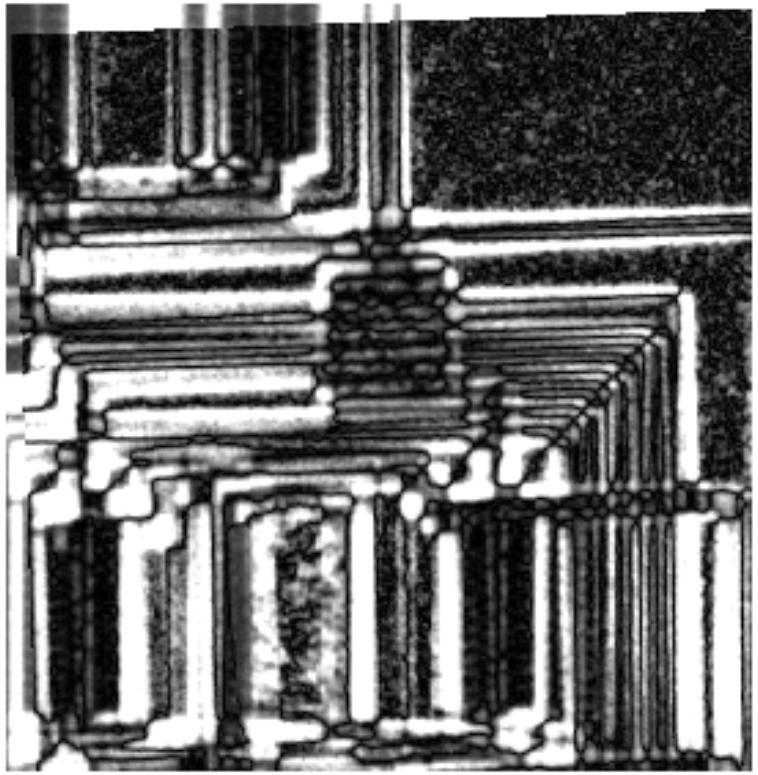


- Start with large diamond pattern at [0,0]
- If best match lies in the center of large diamond, proceed with small diamond
- If best match does not lie in the center of large diamond, center large diamond pattern at new best match

Absolute difference between images

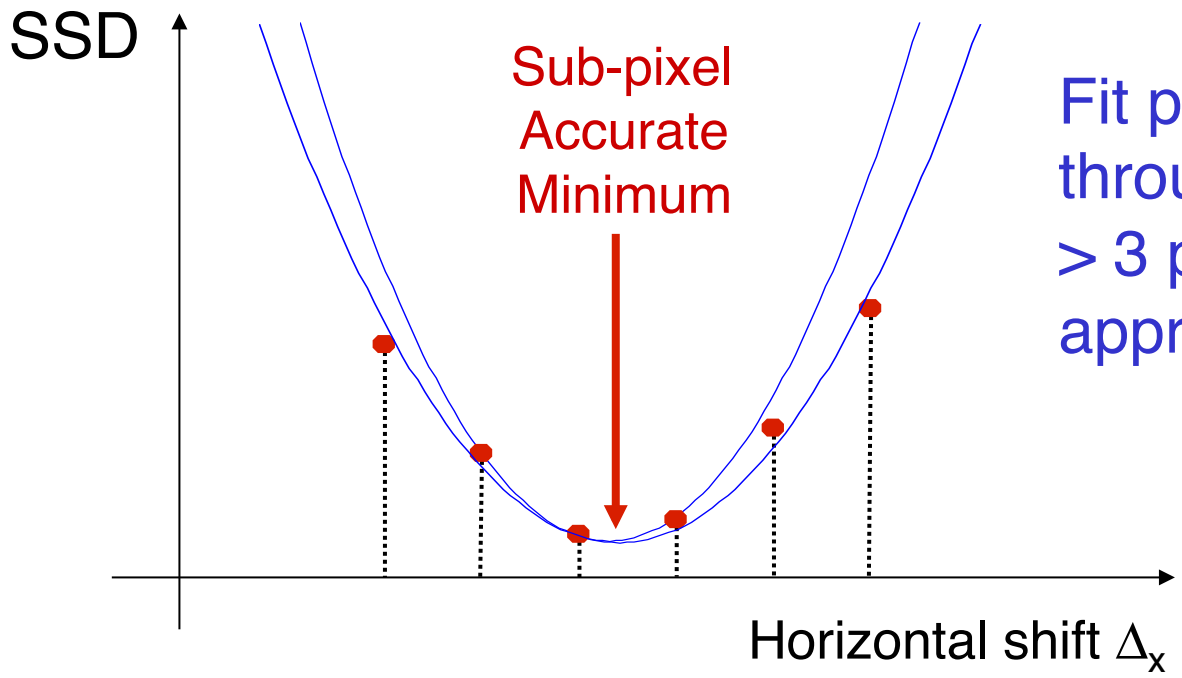


w/ integer-pixel alignment



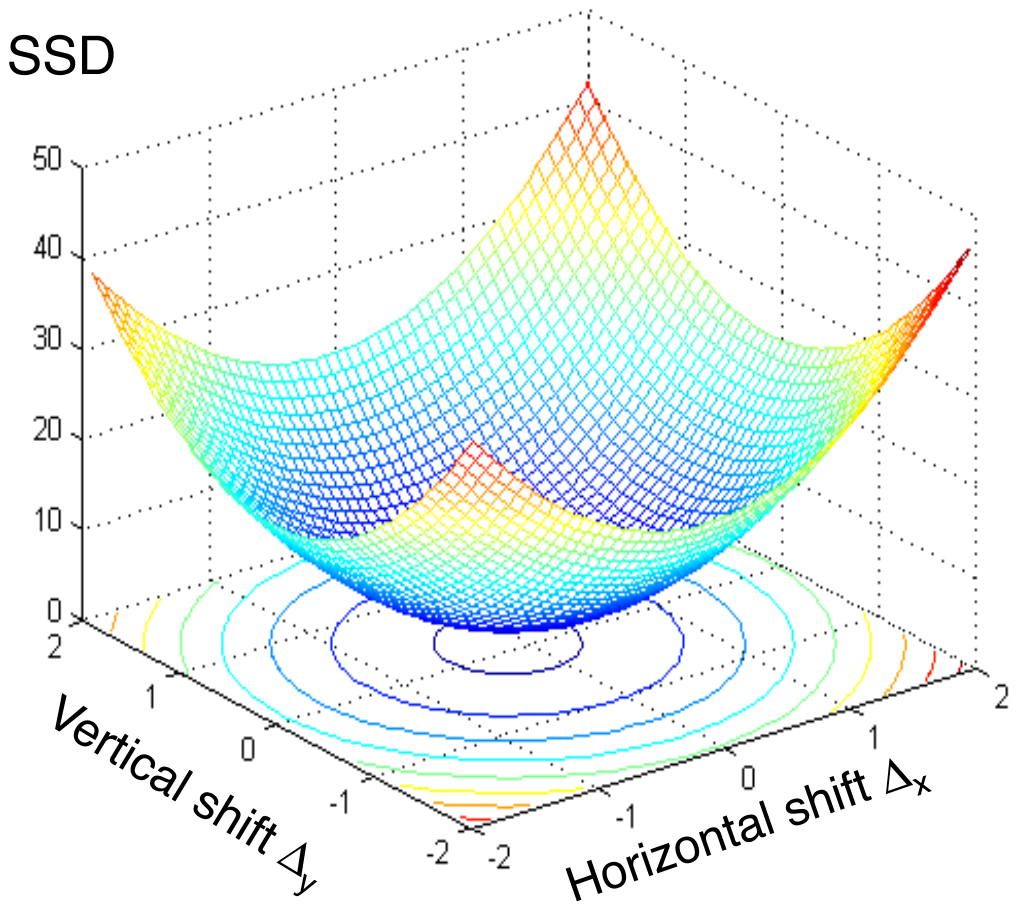
w/o alignment

Interpolation of the SSD Minimum



Fit parabola through > 3 points approximately

2-d Interpolation of SSD Minimum



Paraboloid

- Perfect fit through 6 points
- Approximate fit through > 6 points